

6.2 Bayes Theorem – Notes

1. Problem setting in ML

- Goal: Find the **best hypothesis** $h \in H$ given observed training data D .
 - "Best" means: the **most probable hypothesis** given data (posterior probability).
 - Bayes theorem provides the rule to update beliefs about hypotheses after seeing data.
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2. Notation

- $P(h)$: **Prior probability** of hypothesis h (before seeing data).
 - $P(D)$: **Prior probability of data** D (probability that data occurs regardless of hypothesis).
 - $P(D | h)$: **Likelihood** of observing data D if hypothesis h is true.
 - $P(h | D)$: **Posterior probability** of hypothesis h given data D .
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3. Bayes Theorem

$$P(h | D) = \frac{P(D | h) P(h)}{P(D)}$$

- Posterior = (Likelihood $\times$ Prior) $\div$ Evidence
 - Intuition:
 - Posterior increases with prior belief $P(h)$.
 - Posterior increases with likelihood $P(D|h)$.
 - Posterior decreases if $P(D)$ (the general probability of data) is very high.
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4. MAP Hypothesis (Maximum A Posteriori)

- Definition: hypothesis with the highest posterior probability.

$$h_{MAP} = \arg \max_{h \in H} P(h | D)$$

- Using Bayes theorem:

$$h_{MAP} = \arg \max_{h \in H} P(D | h) P(h)$$

- Here we **drop** $P(D)$ since it is the same for all hypotheses.
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5. ML Hypothesis (Maximum Likelihood)

- Special case: when all hypotheses are **equally likely a priori** (uniform prior).
- Then, only the likelihood matters:

$$h_{ML} = \arg \max_{h \in H} P(D | h)$$

- This is the hypothesis that makes the observed data most probable.
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6. Generality of Bayes Theorem

- Applies beyond ML: works for **any set of mutually exclusive propositions**.
 - Example: $H = \{\text{"sky is blue"}, \text{"sky is not blue"}\}$.
 - In ML, H is usually the hypothesis space (candidate models), and D is training data.
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6.2.1 Example – Medical Diagnosis with Bayes Rule

Problem Setup

Two hypotheses:

1. h_1 : Patient has cancer.
2. h_2 : Patient does not have cancer.

Data: A laboratory test result $T \in \{\text{positive, negative}\}$.

Given probabilities:

- Prior: $P(\text{cancer}) = 0.008$, $P(\text{no cancer}) = 0.992$.
 - Test accuracy:
 - $P(\text{positive} | \text{cancer}) = 0.98$ (true positive rate).
 - $P(\text{negative} | \text{no cancer}) = 0.97$ (true negative rate).
 - Complementary probabilities:
 - $P(\text{negative} | \text{cancer}) = 0.02$.
 - $P(\text{positive} | \text{no cancer}) = 0.03$.
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Bayes Theorem Application

We observe: test result = **positive**.

Posterior probability:

$$P(\text{cancer} | \text{positive}) = \frac{P(\text{positive} | \text{cancer}) \cdot P(\text{cancer})}{P(\text{positive})}$$

where

$$P(\text{positive}) = P(\text{positive} \mid \text{cancer})P(\text{cancer}) + P(\text{positive} \mid \text{no cancer})P(\text{no cancer})$$

Numerical Calculation

- Numerator:

$$P(\text{positive} \mid \text{cancer}) \cdot P(\text{cancer}) = 0.98 \cdot 0.008 = 0.00784$$

- Denominator:

$$P(\text{positive}) = (0.98)(0.008) + (0.03)(0.992) = 0.00784 + 0.02976 = 0.0376$$

- Posterior:

$$P(\text{cancer} \mid \text{positive}) = \frac{0.00784}{0.0376} \approx 0.2085 \text{ } (\approx 21\%)$$

So:

- $P(\text{cancer} \mid \text{positive}) \approx 21\%$.
 - $P(\text{no cancer} \mid \text{positive}) \approx 79\%$.
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Key Takeaways

- Even though the test is **98% accurate**, the **low prior probability (0.8%)** means the posterior probability of cancer is only ~21%.
 - Bayesian reasoning **balances prior knowledge + evidence**.
 - Hypotheses are not fully accepted/rejected, only **updated** in probability.
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New Example – Spam Filter

Problem Setup

Two hypotheses:

1. h_1 : Email is **spam**.
2. h_2 : Email is **not spam**.

Suppose:

- Prior knowledge: 20% of emails are spam.

$$P(\text{spam}) = 0.2, \quad P(\text{not spam}) = 0.8$$

- The word “**lottery**” appears in many spam mails:

$$P(\text{“lottery”} \mid \text{spam}) = 0.6$$

$$P(\text{“lottery”} \mid \text{not spam}) = 0.01$$

Now: A new email arrives containing the word “**lottery**.”

Apply Bayes Theorem

$$P(\text{spam} \mid \text{“lottery”}) = \frac{P(\text{“lottery”} \mid \text{spam}) \cdot P(\text{spam})}{P(\text{“lottery”})}$$

where

$$P(\text{“lottery”}) = P(\text{“lottery”} \mid \text{spam})P(\text{spam}) + P(\text{“lottery”} \mid \text{not spam})P(\text{not spam})$$

Numerical Calculation

- Numerator:

$$0.6 \cdot 0.2 = 0.12$$

- Denominator:

$$(0.6)(0.2) + (0.01)(0.8) = 0.12 + 0.008 = 0.128$$

- Posterior:

$$P(\text{spam} \mid \text{“lottery”}) = \frac{0.12}{0.128} \approx 0.94$$

So:

- Posterior probability of **spam** $\approx$ **94%**.
 - Even though only 20% of emails are spam, the presence of the word “lottery” strongly shifts the belief.
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Practice Problem for Students (Spam Filter)

👉 **Problem Statement:**

Assume:

- $P(\text{spam}) = 0.3$, $P(\text{not spam}) = 0.7$.
- The word "**free**" appears in 40% of spam emails and in 5% of non-spam emails.
- A new email contains the word "free."

Q1: What is the probability that this email is spam?

Q2: How does this compare with the prior 30%?